3rd Sep, 2025 Calculus Quiz 1 [redacted]

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Q1:- Evaluate the limit:-

$$\lim_{x\to4} \frac{4-x}{5-\sqrt{x^2+9}}$$

By directly putting the limit we get $\left(\frac{0}{0}\right)$ form, so ~~we~~ rationalize the denominator.

$$\lim_{x\to4} \frac{4-x}{5-\sqrt{x^2+9}} \times \frac{5+\sqrt{x^2+9}}{5+\sqrt{x^2+9}} = \lim_{x\to4} \frac{(4-x)(5+\sqrt{x^2+9})}{(5)^2-(\sqrt{x^2+9})^2}$$

$$= \lim_{x\to4} \frac{(4-x)(5+\sqrt{x^2+9})}{25-(x^2+9)} = \lim_{x\to4} \frac{(4-x)(5+\sqrt{x^2+9})}{25-x^2-9}$$

$$= \lim_{x\to4} \frac{(4-x)(5+\sqrt{x^2+9})}{16-x^2} = \lim_{x\to4} \frac{\cancel{(4-x)}(5+\sqrt{x^2+9})}{(4+x)\cancel{(4-x)}}$$

Now putting value of limit in $\lim_{x\to4} \frac{5+\sqrt{x^2+9}}{4+x}$ we get:-

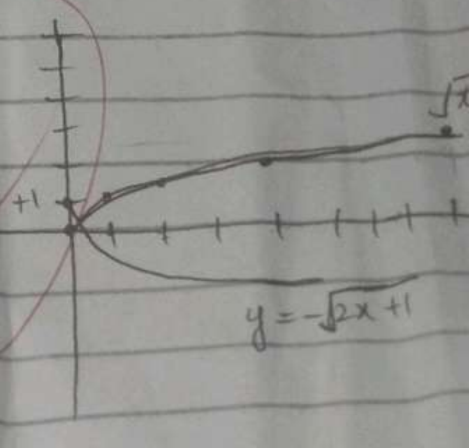
$$= \frac{5+\sqrt{(4)^2+9}}{4+4} = \frac{5+\sqrt{16+9}}{8} = \frac{5+\sqrt{25}}{8} = \frac{5+5}{8} = \frac{10}{8}$$

Hence, $\lim_{x\to4} \frac{4-x}{5-\sqrt{x^2+9}} = \frac{5}{4} = 1.22$

Q2 Sketch the graph using original graph

$y = -\sqrt{2x+1}$

original graph is $\sqrt{x}$



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① $y = \sqrt{2x} \Rightarrow f(cx)$ -

compresses graph horizontally

② $y = \sqrt{2x+1} \Rightarrow f(x)+k$

~~if k~~ here $k = +1$ so graph is shifted upwards by $+1$

③ $y = -\sqrt{2x+1}$

$y = -f(x) \rightarrow$ rotates the graph along horizontal axis

reflect